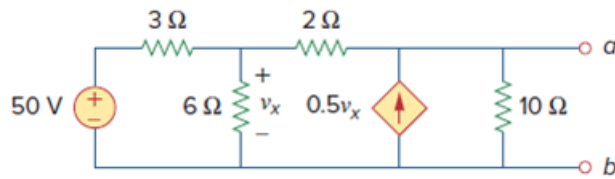


Problem

Obtain the Thevenin and Norton equivalent circuits at terminals $a-b$ for the circuit in Fig. 4.123.

Figure 4.123:



Step-by-step solution

Step 1 of 9

Refer to Figure 4.123 in the textbook.

To determine the Thevenin resistance, all the independent sources in the circuit are deactivated, the voltage source is replaced with a short circuit and the current source is replaced with an open circuit.

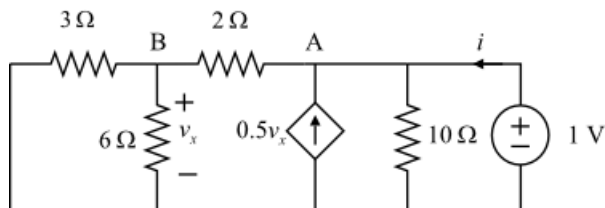
If the circuit contains a dependant source, after deactivating the independent sources place a test source across the terminals across which the Thevenin's resistance is required.

In the circuit under consideration, place a 1 V test source across the $a-b$ terminals as these are the terminals across which the Thevenin's resistance needs to be found.

$$v_a = 1 \text{ V}$$

Let the current leaving the current source be i .

Redraw the circuit with the voltage source short circuited and the test source in between the $a-b$ terminals.



Step 2 of 9

Apply Nodal analysis at the node A.

$$i + 0.5v_x = \left(\frac{1}{10}\right) + \left(\frac{1-v_x}{2}\right)$$

$$i + 0.5v_x = 0.1 + 0.5 - 0.5v_x$$

$$i + 0.5v_x + 0.5v_x = 0.6$$

$$i + v_x = 0.6 \dots\dots (1)$$

Apply Nodal analysis at the node B.

$$\frac{v_x - v_A}{2} + \frac{v_x}{6} + \frac{v_x}{3} = 0$$

$$\frac{v_x - 1}{2} + \frac{v_x}{6} + \frac{v_x}{3} = 0$$

$$\frac{3(v_x - 1) + v_x + 2v_x}{6} = 0$$

$$\frac{6v_x - 3}{6} = 0$$

$$6v_x - 3 = 0$$

$$v_x = \frac{1}{2} \text{ V}$$

$$v_x = 0.5 \text{ V}$$

Substitute 0.5 V for v_x in equation (1).

$$i + 0.5 = 0.6$$

$$i = 0.1 \text{ A}$$

Thevenin's resistance is the ratio of the test source voltage and the current it is generating.

The expression for the Thevenin's resistance is,

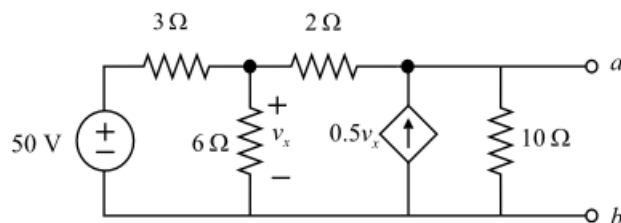
$$\begin{aligned} R_{Th} &= \frac{v_A}{i} \\ &= \frac{1}{0.1} \\ &= 10 \Omega \end{aligned}$$

Thus, the Thevenin's resistance is 10Ω .

Step 3 of 9

The Thevenin's voltage is the voltage across the a - b terminals due to the 50 V voltage source.

Label the nodes and redraw the circuit diagram.



Step 4 of 9

Apply nodal analysis at the node with node voltage v_x .

$$\frac{v_x - 50}{3} + \frac{v_x}{6} + \frac{v_x - v_a}{2} = 0$$

$$\frac{2(v_x - 50) + v_x + 3(v_x - v_a)}{6} = 0$$

$$2v_x - 100 + v_x + 3v_x - 3v_a = 0$$

$$6v_x - 3v_a = 100 \dots\dots (2)$$

Step 5 of 9

Apply nodal analysis at the node a .

$$\frac{v_a - v_x}{2} + \frac{v_a}{10} = 0.5v_x$$

$$\frac{6v_a}{10} = v_x$$

$$\frac{3}{5}v_a = v_x \dots\dots (3)$$

Substitute equation (3) in equation (2).

$$6\left(\frac{3v_a}{5}\right) - 3v_a = 100$$

$$\left(\frac{18v_a - (5(3v_a))}{5}\right) = 100$$

$$\frac{18v_a - 15v_a}{5} = 100$$

$$3v_a = 500$$

$$v_a = 166.67 \text{ V}$$

But the node voltage v_a is the voltage across the terminals a - b or it is the Thevenin's voltage.

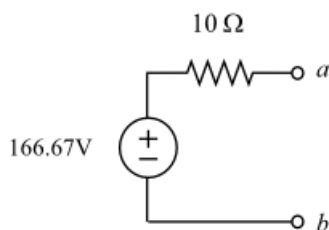
$$v_a = v_{ab} = v_{th}$$

$$v_{th} = 166.67 \text{ V}$$

Thus, the Thevenin's voltage is 166.67 V.

Step 6 of 9

Draw The resultant Thevenin's circuit.



Step 7 of 9

The Norton's resistance is equal to the Thevenin's resistance.

$$\begin{aligned} R_N &= R_{th} \\ &= 10 \Omega \end{aligned}$$

Thus, the Norton's resistance is 10 Ω.

Step 8 of 9

Write the expression for the short circuit current in terms of Thevenin's voltage.

$$I_N = \frac{V_{Th}}{R_{Th}}$$

Substitute $10\ \Omega$ for R_{Th} and 166.67 for V_{Th} .

$$\begin{aligned} I_N &= \frac{166.67}{10} \\ &= 16.67\text{ A} \end{aligned}$$

Thus, the Norton current has a value of 16.67 A.

Step 9 of 9

Draw the resultant Norton's circuit.

